InnovizTwo for Emergency Brake Assist

Lidar performance – statistical modelling (Exercise 3)

Team: *names TBD*

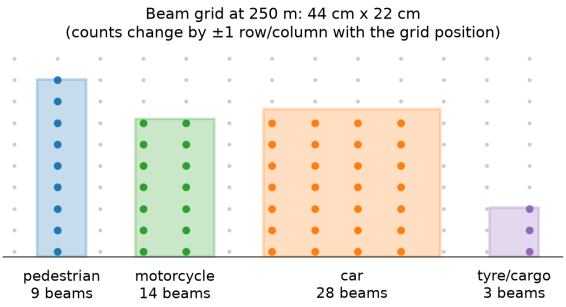
3DSSF – Kozma

The lidar: InnovizTwo (Innoviz)

- automotive-grade, for mass production (L2–L5)
- forward ROI 30° × 9.6°: **0.1°** × **0.05°** resolution
- **210 m @ 10 %** reflectivity, **90 % true positives per pixel** (100 klx sun)
- 120° × 28.8° FoV, 20 FPS, 905 nm

The datasheet gives the per-beam detection probability – the key input of our model

Source: InnovizTwo Series Production Datasheet v3.1 (2024), Condor configuration.



Model: returns per frame are a random variable

1. Beams on target (W × H at R)

$$n = \frac{W}{R\,\Delta\theta_h} \cdot \frac{H}{R\,\Delta\theta_v}$$

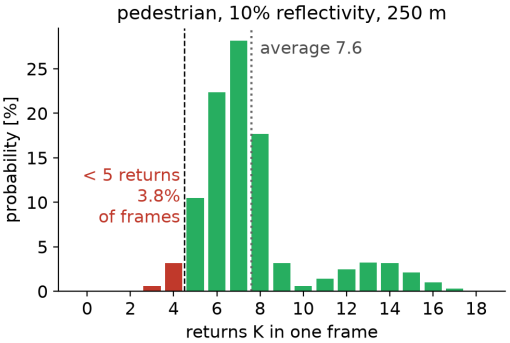
random grid offset → n rounds up or down

2. Each beam returns with prob.

$$p = 1 - e^{-C\rho/R^2}, \quad p(210\,\text{m}, 10\,\%) = 0.9$$

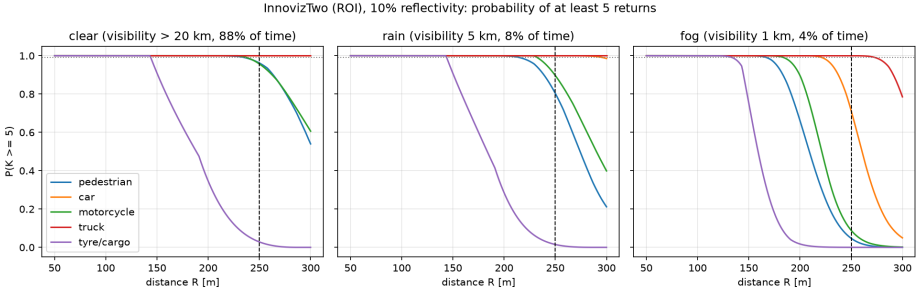
3. Returns: $K \sim \text{Binomial}(n, p)$

Requirement: $P(K \geq 5) \geq 99\,\%$ at 250 m



average $7.6 \geq 5$, but
$P(K \geq 5) = 96.2\,\% < 99\,\%$
⇒ **the average passes, the probability fails**

Results: 10 % reflectivity, three weather conditions



	$P(K \geq 5)$ at 250 m				99 % range [m]		
	clear	rain	fog	**weighted**	clear	rain	fog
car	100 %	100 %	**71 %**	98.8 %	≥300	296	**221**
pedestrian	**96.2 %**	**80 %**	**5 %**	**91.3 %**	**235**	**215**	**167**
motorcycle	**95.7 %**	**90 %**	**9 %**	**91.8 %**	**237**	**232**	**184**

truck passes in all three, tyre fails in all. Weighted: 88 % clear, 8 % rain, 4 % fog (our estimate for Hungary).

Conclusion: EBA met for vehicles, not for dark road users

- cars and trucks: pass at 250 m even at 10 % reflectivity
- dark pedestrians / motorcycles: 99 % only up to ≈ 235 m – 15 m short (bright ones pass)
- small debris: ≈ 140 m – only 2–3 beams hit
- fog (1 km visibility): even cars only to 221 m → adapt speed
- bottleneck: 0.1° horizontal resolution → narrow objects get 1–2 columns

Limitations: flat frontal targets; independent beams; false positives (1 %/pixel) ignored; single frame – at 130 km/h, 68 more frames before braking must start.

